

GIT / GITHUB CRASH COURSE

Master of Science in Machine Learning and Data Science (MLDS)
Class of 2027, Omicron Cohort

Agenda

1. Git Workflow

2. GitHub Workflows

- Create your own GitHub repo (*push + pull*)
- Work with an existing GitHub repo (*clone + fork*)
- Submit an assignment (*pull request*)

Mini Deliverable

- By the end of this afternoon, your task is to submit a pull request to our **NUMLDS/bootcamp-2026** repo

What is git?

It provides version control.

Before git

- You save a completely new version of a file each time you update it (v1, v2, etc.)

After git

- Git tracks what changed between versions of a file
- You can always go back to previous check points and undo / update changes

Fun fact: *git was invented by Linus Torvalds (creator of Linux) in 2005*

EXERCISE: git workflow

1. Open your shell

- Terminal (MacOS)
- Git Bash (Windows)

2. Setup git configurations (one-time only)

- Specify your name and email

3. Go through basic git workflow (every time)

- a. Create a new coding project (repo)
- b. Activate git (turn on version control)
- c. Make an update in the repo
- d. Save it as a checkpoint (make a commit)
- e. View previous checkpoints (versions)

repo = your
coding project

commit = save
a checkpoint

DEMO: git workflow

```
git config --list
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

```
cd Desktop
mkdir my_git_project
cd my_git_project
git init
ls -la
touch README.md
git status
git add README.md
git status
git commit -m "first commit"
git status
git log
```

- **Mac:** Install Xcode Command Line Tools Prompt? Yes.
- **PC:** Git Bash comes with git

Demo: Turn R project from this morning into a git repo

EXERCISE: git workflow

Within the same repo:

1. Make some updates

- Open README.md and type "hello world"
- Create a new file called scratch.txt

2. Go through the git workflow to start tracking changes

- View the **status** of the files
- **Add** the files to staging
- **Commit** the changes
- View the change **log**

SOLUTION: git workflow

```
vim README.md  
i  
hello world  
Esc  
:wq  
touch scratch.txt
```

```
git status  
git add .  
git status  
git commit -m "updated files"  
git status  
git log
```

Practical Git Tips

If you ever mess something up

- Delete the .git folder and do another git init to start over

If you're ever unsure of your git commands skills

- Make a copy of your folder without git as a backup

Be careful what git commands you run (i.e. git reset --hard)

- You could lose edits, so be sure to understand commands before executing

There are many more git commands

- No need to memorize them, you will learn over time with lots of trial and error

Git vs GitHub

Git

- The actual version control tool
- Runs locally on your computer

local = your
computer

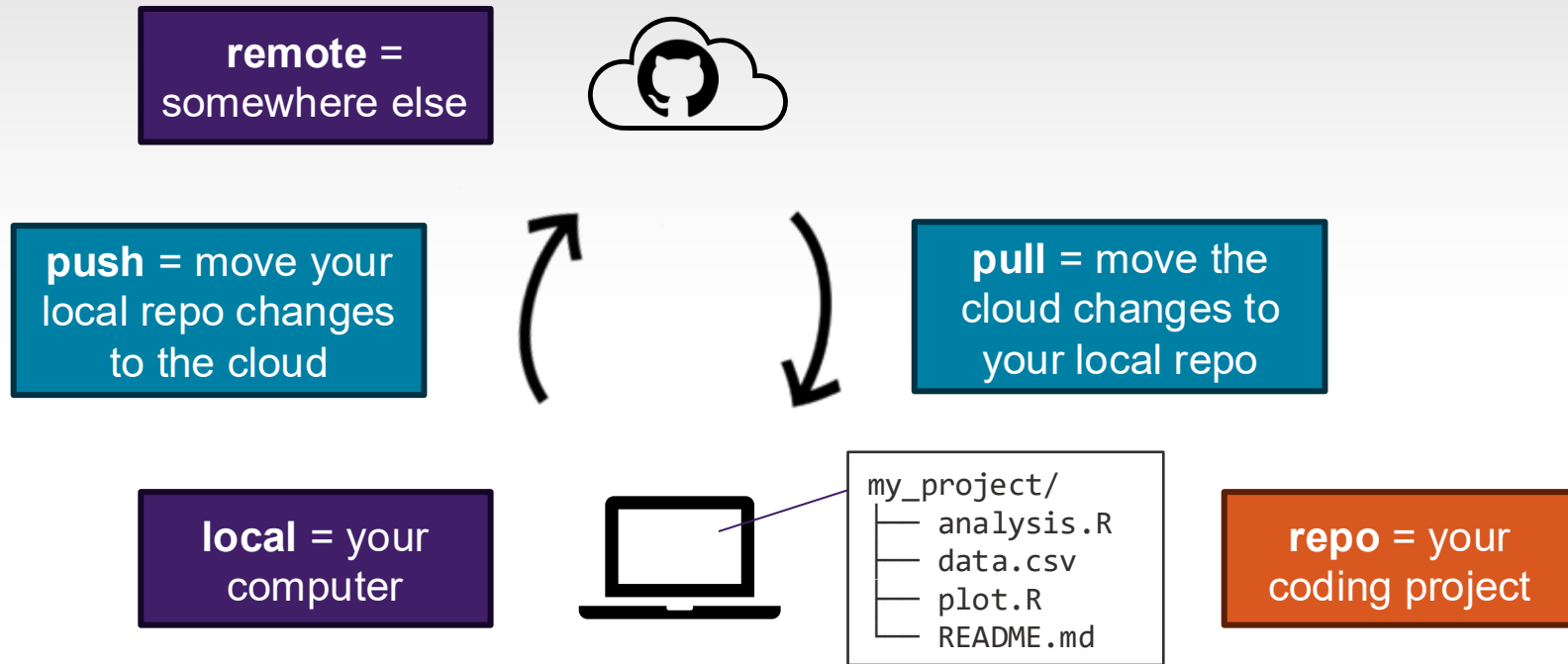
GitHub

- Website that hosts your git repos online, so others can access them
- Acquired by Microsoft in 2018
- Alternatives: GitLab, Bitbucket, etc.

remote =
somewhere else

repo = your
coding project

GitHub Key Terms



QUIZ: Define these terms

- Git
- Github
- Repo
- Add
- Commit
- Log
- Local
- Remote
- Push
- Pull

GITHUB

Getting Started with GitHub

First-time setup (one time)

- Create a GitHub account
- Generate an SSH key

Typical workflow (every time)

- Create your own GitHub repo
- Work with an existing GitHub repo
- Submit a pull request



GitHub: First-Time Setup

1. Create a free GitHub account

- <https://github.com/signup>

2. Generate an SSH key

- <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>

3. OPTIONAL: Sign up for GitHub Education

- <https://github.com/education>

SSH Key Overview

What are the different ways you can interact with a GitHub repo?

- **HTTPS** – need to enter in username and password with every push
- **SSH (preferred approach)** – set up once and you're all set

What is an SSH Key?

- Pair of keys (random alphanumeric numbers) that allows your computer to prove your identity to GitHub
- **Private key** = stays on your computer and is never shared with anyone
- **Public key** = uploaded to your GitHub settings

DEMO: SSH Key Steps

GitHub's official documentation

- <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>

Step 1: Generate a key pair on your computer

Step 2: Add the public key to GitHub

IMPORTANT: If you've already generated a GitHub SSH key in the past, when prompted to overwrite your key, choose **no**.

GITHUB WORKFLOWS

1. Create Your Own GitHub Repo

1. Create a new repo on GitHub: Create new >> New repository

2. Go to the repo folder on your laptop

```
cd my_git_project
git remote -v
git remote add origin git@github.com:<your-username>/my_git_project.git
git remote -v
git push -u origin main
```

3. Make a change locally

```
echo "my first repo" >> README.md
git status
git add README.md
git status
git commit -m "updated readme"
git push
```

4. Make a change on GitHub

```
git pull
cat README.md
```

2a. Work with an Existing Repo (Clone)

Goal: Pull daily updates from the class repo

Option A: Create a separate non-git folder on your laptop

- Point-and-click to copy files from the cloned repo to a folder on your laptop

1. Clone the class repo (one-time)

```
git clone git@github.com:NUMLDS/bootcamp-2026.git  
cd bootcamp-2026
```

2. When new content is added (ongoing)

```
git pull
```

2b. Work with an Existing Repo (Fork)

Goal: Pull daily updates from the class repo

Option B: Create a forked repo

- (One-time) Fork bootcamp-2026 on GitHub first, then:

```
git clone git@github.com:<your-username>/bootcamp-2026.git  
cd bootcamp-2026
```

```
git remote add upstream git@github.com:NUMLDS/bootcamp-2026.git  
git remote -v
```

```
git pull upstream main  
git push origin main
```

3. Submit a Pull Request

1. Go to your forked repo and pull the latest updates from the class repo

```
cd bootcamp-2026  
git pull upstream main
```

2. Add a file to the submissions folder

```
cd submissions  
echo "testing 123" >> lastname_firstname.txt
```

3. Submit a pull request for your forked repo edit to be accepted by the class repo

```
git status  
git add .  
git commit -m "submitted assignment"  
git push origin main
```

4. Within GitHub, click Contribute >> Open pull request, and view the PR in the class repo

Practical GitHub Tips

How to avoid git issues

- They are unavoidable unfortunately – you will encounter them
- Before you start working – git pull
- When you're done working – git push

The git workflow is the same everywhere

- In the command line, GitHub Desktop, VS Code, etc.

There are many additional features within GitHub

- Public vs private repos, teams, automated workflows, etc.

Practical GitHub Tips

Write clear commit messages

- AI helps with this now!

Read error messages

- They usually tell you how to fix your issue

Stuck?

- Run git status to see where you are

Use .gitignore to keep your repos clear of hidden files

NEXT STEPS

GIT / GITHUB CRASH COURSE

SUMMARY

-  Git is just a way to save checkpoints of your work, so you can always go back
-  GitHub is a website that hosts your repos online, so you can code with others
-  You now know the solo git workflow -- later during fall quarter we'll discuss group workflows including pull requests, merge conflicts, branches, etc.

NEXT STEPS

-  Fill out the MLDS student tech info spreadsheet (from Mateusz)
-  [Optional] Sign up for GitHub Education (<https://github.com/education>)

SMALL GROUP DISCUSSION

We've completed our git and GitHub crash course! 🎉

Discuss in groups of 3-4 people:

- How do you feel about git / GitHub?
- What are some questions you still have about git / GitHub?
- What's something new you learned today?